

EVENTIM Architecture Manifesto

The 12 Principles

At EVENTIM, we're not just writing code - we're building the foundation for unforgettable live experiences around the world. Our architecture shapes how fast we move, how well we scale, and how boldly we innovate. That's why we've come together to define this Architecture Manifesto - a shared vision for how we design, build, and grow our systems.

Let's do it right. Let's do it together.

Platform Approach

We believe in building once — and building well. By abstracting common **capabilities** into **reusable, shared services**, we enable teams to **develop faster**, stay aligned, and deliver consistent, scalable solutions across the entire EVENTIM ecosystem.

? Example: Instead of each team implementing their own user authentication logic, a central Auth Service provides OIDC, SSO, and MFA — reused across Ticketing, App, and Promoter Portal.

KPIs Service Reuse Ratio Platform Ownership Clarity Cost per Integration

Internationalization and Multi-Tenancy

We build for the world. That means supporting multiple **languages, regions, and cultural norms** — and doing it all within architectures that serve many tenants with the same reliability, flexibility, and scale.

? Example: The CMS allows content editors to manage translations for different markets while the backend dynamically applies tenant-specific themes and pricing models per region.

KPIs Localization Coverage Per-Tenant Configuration Flexibility Scalability Across Tenants

Design for Change

Change is constant — our systems are built to embrace it. We design for **evolution** from day one, so that features, components, and infrastructure can adapt over time with **minimal disruption** and maximum responsiveness to business needs.

? Example: A pricing engine is built with plug-in support for different calculation strategies, enabling easy adjustments for promotions, territories, or business rules without rewriting core logic.

KPIs Hot-Swappability Score Legacy Component Reduction Rate Independent Deployability Rate

Continuous Improvement & Modernization

We build for today — and refactor for tomorrow. By embracing continuous improvement, we keep our systems **lean, modern**, and easy to change. We manage technical debt deliberately and invest in refactoring as a way to stay fast, reliable, and **future-ready**.

? Example: Legacy components are progressively rewritten using modern architecture. Weekly tech-debt standups prioritize outdated libraries and untyped interfaces for refactoring.

KPIs Technical Debt Ratio Refactoring Commitment Rate Retrospective Action Rate

API-First

Clear contracts build strong systems. We define APIs up front — not as an afterthought — to drive **autonomy, reusability, and integration**. Thoughtful versioning and documentation make collaboration seamless and scalable.

? Example: Before developing a new fan loyalty service, the API is defined in an OpenAPI spec. Teams develop in parallel using mocks, accelerating integration and reducing surprises.

KPIs API Re-use Rate Time to First Successful Call Version Stability Ratio

Event-Driven Architecture

We think in signals, not silos. By using events as the backbone of communication, we **decouple** our systems and increase **responsiveness**. **New subscribers?** No problem. Growth is built in.

? Example: When a user books a ticket, an event is published. Other services — like Notifications or Analytics — subscribe and react independently.

KPIs Event Delivery Latency Event Processing Success Rate Event-Driven Service Coverage

AI-Driven

We use AI to work **smarter**, not just faster. From internal optimization to customer-facing innovation, AI helps us automate the routine, uncover insights, and enhance every experience. We design our systems to use AI in a **safe and ethical** way — so we can spend less time on grunt work.

? Example: A recommendation engine uses AI to suggest events based on user behavior. AI also helps operations by forecasting traffic spikes, improving resource allocation proactively.

KPIs Hot-Swappability Score Legacy Component Reduction Rate Independent Deployability Rate

Loose Coupling and High Cohesion

We build modular systems that do one thing well — and work together even better. With loose coupling and high cohesion, each component is **focused, replaceable**, and designed to evolve **independently**, making the whole stronger and more agile.

? Example: Ensure parts of the software interact via well-defined contracts, like APIs, Business Events or software interfaces such as IShippingCalculator.

KPIs Service Dependency Ratio Independent Deployment Rate Team-to-Service Ownership Ratio

Security by Design & Security by Default

Security isn't bolted on — it's built in. From **zero-trust** principles to encryption and **least-privilege access** - following the **need-to-know** principle - we embed security at every level and start with **secure defaults**. Every service protects what matters, by design.

? Example: Zero-trust policies require re-authentication before accessing sensitive customer interactions like changing the account email address in the EVENTIM webshop.

KPIs Security Architecture Compliance Rate Patch Management Latency Security Test Automation Rate

Observability and Monitoring

We don't just build — we watch, learn, and improve. From **logs, metrics and traces** to KPIs, we design for deep visibility into system health and business insights, so teams can act fast and stay ahead.

? Example: Each service emits structured logs, provides key metrics (latency, error rate, throughput), and sends alerts to Teams when anomalies are detected.

KPIs Alert Noise Ratio MTTA (Mean Time to Acknowledge) MTDD (Mean Time to Detect)

Scalability & Resilience

Our systems don't buckle under pressure — they rise to meet it. We architect for graceful growth, **scaling** horizontally or vertically as needed, with **zero downtime** and full confidence. Resilience ensures we stay strong, **recover gracefully** from failure, and continue delivering even when things go wrong.

? Example: During high-demand onsales, autoscaling ensures ticketing services expand horizontally to handle the surge, while a circuit breaker protects from cascading failure.

KPIs Service Availability (Uptime) Peak Load Throughput Elasticity Lag

Cloud Ready

We build for where we are — and where we're going. With **infrastructure as code, automation**, and **containerization**, our systems are ready for the cloud, hybrid or on-premises — always **optimized** for cost, performance and business value.

? Example: Services are containerized with Docker, orchestrated via Kubernetes and deployed on a cloud, on-premises or hybrid. Autoscaling rules help manage peak traffic cost-effectively.

KPIs Containerization Coverage Cloud Cost Efficiency IaC Adoption Rate